1. Project Description: I made a Python-based terminal game where players have to guess a hidden number between 1 and 100. There are three different difficulties, which are Hard(10 attempts), Medium(20 attempts), and Easy(unlimited attempts). The game tracks your attempts and previous guesses and gives you suggestions if your guess is higher or lower than the original number. The game also stores your scores in a separate text file called leaderboard. Lastly, the game will also show you the leaderboard in proper order after each game.

2. Approach and logic used: Initially, I made a simple number guessing game that just tells you if your guess is “Too High” or “Too Low”. Then I moved on to adding different difficulties and made loops and try-except statements to ensure the user input is correct. I implemented the replay system, which allows the user to start a new session after finishing. Following this, I created a writing system that stores each participant's username and score in a text file to prevent the data from being erased after a single session. From there, I made the leaderboard system to show the current leaderboard in proper order, from highest score to lowest (the lower the number of guesses, the better). Lastly, I tested some edge cases and added comments to my code to help others understand its functionality.

3. Data Structures: I used various data structures, such as lists and dictionaries, to store and validate data. For example, I used a dictionary to store the attempts allowed for each difficulty, with the keys being the different levels of difficulty and the values being the number of attempts. I also used lists to store the previous guesses of the user, which I used to check if the user is making multiple guesses of the same number and warn them about it (preventing them from wasting their guesses).

4. Challenges Faced: For most of the time, I was able to understand and implement the project properly. However, when it came to adding and retrieving data from the leaderboard file, I had a lot of issues. I had to go through previous lectures to refresh my memory on adding and retrieving data from a txt file. On top of that, I did not know how to sort the guesses in proper order. Which I solved by going to the office hours. The TAs helped me understand how the sort function works and also helped me deal with some edge cases.

5. Possible Improvements: In the future, I can add better UI by using the Pygame library. On top of that, I can also give the player options to customize the number of attempts and the guessing range on their own. Lastly, I can also refactor the code into a Class-based structure. This will not only make the code easier to debug and simpler to expand, but also add new features such as multiplayer support.